

Comments on Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

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Introduction

As a pediatrician with nearly thirty years of experience caring for children and adolescents with complex medical needs, I find this work both challenging and rewarding. Electronic health records and improved interoperability offer greater access to clinical decision support, enabling providers to make timely and accurate decisions while reducing patient risk. Recent advances in artificial intelligence make these improvements increasingly possible within my lifetime.

Background

Recent research has reinforced earlier findings that children with complex chronic conditions (CCCs)—defined as illnesses persisting for over twelve months, requiring specialist care, and frequently necessitating hospitalization—remain a smaller yet expanding demographic. Although the prevalence of children with multiple CCCs comprises less than 1% of the pediatric population, this proportion is increasing. In 2022, children with one or more CCCs accounted for 21.9% of hospital discharges, 44.1% of bed days, and 59.5% of hospital charges, marking a substantial rise since 2000, when figures were 16.9%, 32.0%, and 44.2%, respectively. (1)

Children with multiple CCCs were identified as the primary contributors to increased use of hospital resources. Additionally, there was a notable rise in technology dependence—particularly gastrointestinal-related technologies—and comorbid mental health disorders, especially neurodevelopmental conditions, among hospitalized children with CCCs. These results indicate that the population of hospitalized children with CCCs has become increasingly medically complex over time.

Analysis

The rising number of children and adolescents with complex chronic conditions (CCC) is increasing healthcare costs. While AI is widely used for adult chronic diseases to enhance patient engagement, prevent illness, and manage disease, there is limited focus on pediatric models.

Children with complex medical needs require age-specific models addressing their physiology, medication dosing, and unique disease presentations. The American Academy of Pediatrics advises that AI models for pediatrics should be trained and tested using relevant pediatric data to avoid bias.[2]

Distinct challenges exist for this population, including developmental differences, fewer significant clinical outcomes, smaller datasets, and ongoing issues with data sharing, all of which slow advances in pediatric AI models.

Recommendations

Federated learning enables institutions, both small and large, to collaboratively train models while keeping patient data private, making it ideal for national use and overcoming limited pediatric datasets. Institutions can contribute without sharing sensitive information, which is crucial for protecting minors' privacy.

A single, comprehensive language model for all pediatric use cases would be challenging to develop, validate, and maintain. Instead, we propose organ-system–specific models (e.g., respiratory, neurologic, gastrointestinal), each wrapped in a dedicated agent responsible for its safety, governance, and continuous improvement.

Conclusions

By securely combining data from diverse pediatric institutions, these models can capture patterns that no single center could detect alone, enabling more accurate and context-aware clinical decision support. As model performance improves, clinicians are better equipped to make timely, individualized care decisions, which in turn has the potential to enhance patient outcomes, reduce unwarranted variation, and streamline workflows. In aggregate, this shared-data approach supports more efficient care delivery by decreasing diagnostic uncertainty, minimizing redundant testing, and focusing clinicians' time on the highest-value interventions.

Given the limited size and fragmentation of pediatric datasets, no single institution can independently develop robust, generalizable AI models for children. To address this, a coordinated public–private initiative is needed that provides shared technical infrastructure, common governance standards, and sustainable funding for federated learning across pediatric centers. Such an effort should explicitly lower barriers for small and resource-constrained hospitals while also engaging large academic institutions, ensuring that all participants can contribute data and expertise without relinquishing local control or compromising patient privacy. By aligning regulatory support, financial incentives, and standardized participation frameworks, this collaboration would create a more equitable and effective national ecosystem for developing and maintaining pediatric AI models.

References

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